

Gonzalo Benegas

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Summary

ML researcher with a PhD in Computational Biology. Deep expertise in genomic language models, broadly interested in applied ML research.

Experience

Open Athena, Research Scientist – New York, NY 2025 – present

- Leading an open research program on genomic language models.
- Developing scalable data curation strategies and training autoregressive transformers, leveraging NLP infrastructure and scaling recipes.

University of California, Berkeley, Postdoctoral Researcher, EECS – Berkeley, CA 2024 – 2025

- Co-developed GPN-Star, a phylogeny-informed genomic language model achieving SOTA variant effect prediction across both rare and common variants in the human genome (bioRxiv, 2025).
- Co-authored a review on genomic language models, featured on the cover of Trends in Genetics (2025).
- Created TraitGym, a benchmark for evaluating DNA sequence models on variant effect prediction (bioRxiv, 2025). Used as evaluation benchmark by DeepMind's AlphaGenome (Nature, 2026).

University of California, Berkeley, Graduate Student Researcher – Berkeley, CA 2018 – 2023

- Developed GPN-MSA, the first alignment-based genomic language model, achieving SOTA variant effect prediction for rare variants and scalable inference across all 9B possible SNVs in the human genome (Nature Biotechnology, 2025).
- Developed GPN, the first genomic language model to exhibit zero-shot variant effect prediction capabilities across the entire genome (PNAS, 2023).
- Developed scQuint, a variational auto-encoder for analysis of alternative splicing in single-cell RNA-seq (eLife, 2022).

Autodesk, Software Engineer – Buenos Aires, Argentina 2015 – 2016

Syntheticity, Software Engineer – Buenos Aires, Argentina 2014 – 2015

Education

PhD **University of California, Berkeley**, PhD in Computational Biology – Berkeley, CA 2023

- Thesis: Computational and Machine Learning Methods for Understanding Gene Regulation and Variant Effects
- Advisor: Yun S. Song

BS+MS **University of Buenos Aires**, BS+MS in Computer Science – Buenos Aires, Argentina 2018

- GPA: 9.70/10
- Thesis: Bayesian Analysis of Transcriptional Dynamics
- Advisors: Alejo Salles and Hernan G. Garcia

Publications

Mapping cis-regulatory mutations at scale in sorghum enables modulation of gene expression 2026

Evan D. Groover, David Ding, Flora Z. Wang, **Gonzalo Benegas**, Joseph Rivera, Shahar Schwartz, Stephen Chen, Michael F. Moubarak, Viktoriya Georgieva, Peggy G. Lemaux, Brian J. Staskawicz, Krishna K. Niyogi, Yun S. Song, David F. Savage (Nature Biotechnology)

Predicting functional constraints across evolutionary timescales with phylogeny-informed genomic language models 2025

Chengzhong Ye†, **Gonzalo Benegas**†, Carlos Albors, Jianan Canal Li, Sebastian Prillo, Peter D. Fields, Brian Clarke, Yun S. Song (bioRxiv)

A DNA language model based on multispecies alignment predicts the effects of genome-wide variants	2025
Gonzalo Benegas , Carlos Albors, Alan J. Aw, Chengzhong Ye, Yun S. Song (Nature Biotechnology, 43, 1960–1965)	
Genomic language models: opportunities and challenges	2025
Gonzalo Benegas [†] , Chengzhong Ye [†] , Carlos Albors [†] , Jianan Canal Li [†] , Yun S. Song (Trends in Genetics, 41(4), 286–302)	
Benchmarking DNA Sequence Models for Causal Regulatory Variant Prediction in Human Genetics	2025
Gonzalo Benegas , Gökçen Eraslan, Yun S. Song (bioRxiv)	
A Phylogenetic Approach to Genomic Language Modeling	2025
Carlos Albors, Jianan Canal Li, Gonzalo Benegas , Chengzhong Ye, Yun S. Song (RECOMB 2025, pp. 99–117)	
DNA language models are powerful predictors of genome-wide variant effects	2023
Gonzalo Benegas , Sanjit Singh Batra, Yun S. Song (Proceedings of the National Academy of Sciences, 120(44), e2311219120)	
choros: correction of sequence-based biases for accurate quantification of ribosome profiling data	2023
Amanda Mok, Robert Tunney, Gonzalo Benegas , Edward W. J. Wallace, Liana F. Lareau (bioRxiv)	
Robust and annotation-free analysis of alternative splicing across diverse cell types in mice	2022
Gonzalo Benegas , Jonathan Fischer, Yun S. Song (eLife, 11, e73520)	

Presentations

Invited talk, Kipoi Seminar – Virtual	2025
Invited talk, Google Research Genomics Seminar – Virtual	2025
Invited talk, Variant Effects Seminar Series – Virtual	2024
Poster, Machine Learning in Computational Biology (MLCB) – University of Washington	2024
Talk, AI=Science workshop, Simons Institute for the Theory of Computing	2024
Invited talk, Debora Marks Lab, Harvard	2023
Invited talk, iGEM at Berkeley Journal Club – UC Berkeley	2023
Talk, Machine Learning in Computational Biology (MLCB) – Virtual	2022
Invited talk, Moi Exposito-Alonso Lab, Stanford	2022
Invited talk, Luca Pinello Lab, Harvard	2022
Talk, Center for Computational Biology Annual Retreat – UC Berkeley	2021
Poster, CZI Seed Networks 2020 Annual Meeting – Chan Zuckerberg Initiative	2020

Teaching

UC Berkeley , Graduate Student Instructor BIOENG C131: Introduction to Computational Molecular and Cell Biology	2021
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Awards

Reviewers' Choice Poster, ASHG Annual Meeting – Washington, DC	2023
Best poster, runner-up, Center for Computational Biology Annual Retreat – UC Berkeley	2022
Secondary School Exchange Scholar, English-Speaking Union	2011
Second Top Score in Argentina, Cambridge AICE Diploma	2010
Top Score in Argentina, Cambridge AS Level Mathematics	2010
Top Score in Argentina, Cambridge AS Level Biology	2010
Honorable Mention, Argentine National Mathematical Olympiad	2010
Champion, Buenos Aires Province Mathematical Olympiad	2010
Top Score in Argentina, Cambridge IGCSE Chemistry	2009
Top Score in Argentina, Cambridge IGCSE Biology	2009
Champion, Buenos Aires Province Mathematical Olympiad	2009

Service

Reviewer, Nature Methods	2025
Reviewer, Genome Biology	2025
Reviewer, Theoretical and Applied Genetics	2025
Reviewer, Cal Undergraduate BioE Symposium, UC Berkeley	2023
Co-organizer, Center for Computational Biology Student Seminar, UC Berkeley	2021 – 2023
Co-organizer, Center for Computational Biology Seminar, UC Berkeley	2018 – 2019
Questioning Panel Member, National Public Speaking Competition, English-Speaking Union	2014